

## RUSSELL, KS, USA

WMO: 724585

Lat: 38.876N

Lon: 98.809W

Elev: 1864

StdP: 13.73

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93997

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 2.6        | 7.4 | -5.1                          | 4.5 | 5.2  | -0.3 | 5.8 | 10.3 | 33.5                  | 38.2 | 28.9 | 38.1 | 10.6                  | 330  | 0.674 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.2                   | 100.6           | 72.2 | 97.5 | 72.2 | 94.3 | 72.1 | 76.5                | 90.6 | 75.2 | 89.5 | 74.0 | 88.3 | 17.6                 | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 72.9                            | 131.0 | 81.9 | 71.6 | 125.1 | 80.6 | 70.1 | 118.9 | 79.8 | 41.2          | 90.3 | 40.0 | 89.4 | 38.8 | 88.0 | 87.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |      |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |      |
| Min               | Max  | Min  | Max                        | Min   | Max  | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)  | (h)                | (i)                                                | (j)       | (k)   | (l)        | (m)   | (n)        | (o)   | (p)        |      |
| 31.2              | 27.3 | 24.8 | -4.4                       | 105.4 | 5.3  | 3.4                | -8.2                                               | 107.9     | -11.3 | 109.9      | -14.3 | 111.8      | -18.1 | 114.3      |      |
|                   |      |      | DB                         | -5.1  | 78.9 | 5.1                | 2.4                                                | -8.8      | 80.7  | -11.8      | 82.0  | -14.7      | 83.4  | -18.4      | 85.1 |
|                   |      |      | WB                         |       |      |                    |                                                    |           |       |            |       |            |       |            |      |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 55.1   | 30.4  | 33.8  | 44.0  | 53.6  | 64.1  | 75.2  | 80.3  | 77.7  | 69.6  | 56.3  | 43.2  | 32.1  |
|                                                                   | DBStd       | 19.96  | 10.90 | 12.30 | 11.65 | 9.82  | 8.70  | 7.29  | 6.15  | 5.88  | 8.75  | 9.60  | 9.86  | 10.42 |
|                                                                   | HDD50       | 2255   | 609   | 465   | 253   | 72    | 4     | 0     | 0     | 0     | 2     | 46    | 246   | 558   |
|                                                                   | HDD65       | 5121   | 1073  | 873   | 652   | 360   | 127   | 9     | 1     | 1     | 52    | 299   | 654   | 1020  |
|                                                                   | CDD50       | 4130   | 1     | 12    | 67    | 179   | 440   | 757   | 938   | 860   | 591   | 240   | 42    | 3     |
|                                                                   | CDD65       | 1522   | 0     | 0     | 2     | 17    | 98    | 316   | 474   | 396   | 191   | 28    | 0     | 0     |
|                                                                   | CDH74       | 17526  | 0     | 6     | 63    | 282   | 1103  | 3647  | 5770  | 4279  | 2000  | 358   | 18    | 0     |
|                                                                   | CDH80       | 8795   | 0     | 1     | 9     | 83    | 427   | 1841  | 3234  | 2186  | 900   | 113   | 1     | 0     |
| Wind                                                              | WSAvg       | 12.4   | 11.5  | 12.3  | 13.5  | 14.5  | 12.8  | 12.9  | 11.8  | 11.0  | 12.2  | 12.6  | 12.4  | 11.6  |
| Precipitation                                                     | PrecAvg     | 26.3   | 0.6   | 0.8   | 2.1   | 2.6   | 3.7   | 3.5   | 3.5   | 3.3   | 2.8   | 1.7   | 1.0   | 0.8   |
|                                                                   | PrecMax     | 40.2   | 1.9   | 2.8   | 9.3   | 7.9   | 10.8  | 9.4   | 11.5  | 7.5   | 12.0  | 4.9   | 4.6   | 3.2   |
|                                                                   | PrecMin     | 12.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.1   | 0.5   | 0.2   | 0.7   | 0.1   | 0.2   | 0.0   | 0.0   |
|                                                                   | PrecStd     | 6.5    | 0.5   | 0.7   | 1.9   | 1.6   | 2.3   | 2.3   | 2.5   | 1.7   | 2.4   | 1.3   | 1.0   | 0.7   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 66.2  | 72.5  | 80.6  | 88.1  | 95.2  | 102.7 | 105.4 | 101.8 | 98.9  | 89.8  | 76.6  | 64.7  |
|                                                                   |             | MCWB   | 47.3  | 51.5  | 56.9  | 61.8  | 67.3  | 71.6  | 72.7  | 71.9  | 70.4  | 65.3  | 56.7  | 50.4  |
|                                                                   | 2%          | DB     | 58.0  | 65.0  | 74.7  | 81.6  | 89.2  | 97.8  | 102.0 | 98.4  | 93.5  | 82.4  | 70.4  | 58.0  |
|                                                                   |             | MCWB   | 44.1  | 49.4  | 56.2  | 60.5  | 68.0  | 71.9  | 72.6  | 72.5  | 70.2  | 62.6  | 53.6  | 46.3  |
|                                                                   | 5%          | DB     | 52.1  | 58.3  | 69.0  | 76.1  | 84.7  | 94.0  | 98.7  | 95.0  | 89.5  | 77.4  | 64.6  | 53.0  |
|                                                                   |             | MCWB   | 41.6  | 46.2  | 53.3  | 59.1  | 66.8  | 71.8  | 72.7  | 72.6  | 69.3  | 60.4  | 51.4  | 43.1  |
|                                                                   | 10%         | DB     | 46.5  | 52.5  | 63.4  | 71.1  | 80.3  | 90.4  | 94.9  | 91.4  | 85.2  | 72.8  | 59.9  | 47.7  |
|                                                                   |             | MCWB   | 38.8  | 43.0  | 52.1  | 56.9  | 65.5  | 71.0  | 72.9  | 72.1  | 67.6  | 59.2  | 49.4  | 40.2  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 49.6  | 55.1  | 63.1  | 66.4  | 73.3  | 77.2  | 78.5  | 78.1  | 74.9  | 70.2  | 60.1  | 54.9  |
|                                                                   |             | MCDB   | 60.8  | 65.8  | 72.5  | 78.8  | 85.3  | 92.1  | 92.5  | 91.2  | 87.6  | 81.5  | 68.8  | 59.5  |
|                                                                   | 2%          | WB     | 45.2  | 51.0  | 59.7  | 63.9  | 71.3  | 75.2  | 76.7  | 76.2  | 72.8  | 66.7  | 56.9  | 47.7  |
|                                                                   |             | MCDB   | 56.4  | 61.6  | 67.1  | 75.8  | 83.1  | 90.3  | 91.1  | 89.7  | 86.6  | 76.1  | 65.2  | 56.5  |
|                                                                   | 5%          | WB     | 42.2  | 47.0  | 56.3  | 61.2  | 69.2  | 73.6  | 75.4  | 74.8  | 71.2  | 63.9  | 53.3  | 43.7  |
|                                                                   |             | MCDB   | 51.3  | 58.1  | 66.2  | 72.1  | 80.5  | 88.2  | 90.0  | 88.3  | 85.1  | 73.4  | 62.2  | 52.4  |
|                                                                   | 10%         | WB     | 39.1  | 43.5  | 52.1  | 58.4  | 67.3  | 72.3  | 74.3  | 73.4  | 69.7  | 60.9  | 50.2  | 40.6  |
|                                                                   |             | MCDB   | 46.5  | 52.7  | 63.0  | 69.8  | 77.7  | 86.2  | 89.0  | 86.8  | 81.8  | 70.1  | 59.9  | 47.6  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 22.1  | 23.0  | 25.0  | 24.8  | 23.4  | 23.9  | 24.2  | 23.0  | 23.6  | 24.8  | 24.3  | 21.3  |
|                                                                   |             | MCDBR  | 32.2  | 33.3  | 34.0  | 32.7  | 29.0  | 28.4  | 29.0  | 27.5  | 27.4  | 31.3  | 31.9  | 29.5  |
|                                                                   |             | MCWBR  | 20.1  | 19.6  | 18.1  | 16.4  | 12.5  | 9.4   | 8.2   | 9.1   | 9.9   | 14.8  | 17.8  | 18.8  |
|                                                                   | 5% WB       | MCDBR  | 30.5  | 31.4  | 28.2  | 27.6  | 24.8  | 25.3  | 24.5  | 23.7  | 23.8  | 24.5  | 27.9  | 27.7  |
|                                                                   |             | MCWBR  | 19.5  | 19.3  | 16.7  | 15.7  | 12.5  | 10.2  | 9.0   | 9.8   | 9.8   | 13.5  | 17.1  | 18.5  |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.273 | 0.282 | 0.315 | 0.342 | 0.369 | 0.389 | 0.409 | 0.396 | 0.361 | 0.318 | 0.281 | 0.270 |
|                                                                   | taud        |        | 2.527 | 2.502 | 2.433 | 2.388 | 2.394 | 2.369 | 2.343 | 2.378 | 2.415 | 2.516 | 2.545 | 2.527 |
|                                                                   | Ebn at Noon |        | 291   | 303   | 300   | 296   | 289   | 282   | 275   | 276   | 281   | 285   | 286   | 284   |
|                                                                   | Edh at Noon |        | 23    | 28    | 33    | 37    | 37    | 38    | 39    | 37    | 34    | 27    | 23    | 22    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 781   | 1037  | 1391  | 1730  | 1995  | 2234  | 2184  | 1934  | 1596  | 1165  | 866   | 680   |
|                                                                   | RadStd      |        | 64    | 105   | 109   | 96    | 130   | 120   | 108   | 113   | 95    | 121   | 72    | 58    |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (21 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page